



Projective measurements are equilibration processes

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Abstract

We characterize a projective measurement as a black-box process that returns the a priori post-measurement ensemble. We show that this process can be understood as equilibration under a Lindblad equation or as selection of the closest ensemble, in terms of Hilbert–Schmidt distance, commuting with the measured observable. If we restrict to ensembles with finite entropy, closeness can be expressed through the relative entropy.

1 Introduction

Projective measurement is usually presented together with state update: one measures an observable, obtains a particular value and assigns the corresponding conditional state. This combines two different questions. The first is the prediction of the outgoing ensemble given only the incoming ensemble and the measured observable. The second is the incorporation of the measurement outcome once it is known. Here we consider only the first.

We treat measurement as a black-box process. The description refers only to incoming and outgoing ensembles and makes no claim about the internal mechanism by which the measurement is implemented. From this point of view, the usual projective-measurement prediction can be characterized in four ways. It is the standard pinching of the incoming ensemble, the equilibrium reached under a dephasing Lindblad equation and the closest ensemble compatible with the measured observable, either through the Hilbert–Schmidt distance or the relative entropy.

2 Conditions

The following condition represents the standard structure of quantum states and observables on Hilbert space.

Condition QSTO:H (Quantum states and observables). *The state space of a quantum system is represented by the projective space $\mathcal{P}(\mathcal{H})$ of a separable complex Hilbert space $\mathcal{H}$. The ensemble space is represented by the density operators $\mathcal{D}(\mathcal{H})$. Observables are represented by self-adjoint operators. The conditional probability is given by the Born rule $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$. The entropy is given by the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$.*

This base condition states the ensembles have finite entropy.

Condition [B]FINENT (Entropy is finite). *All ensembles have a well-defined finite entropy. Therefore, mathematical objects representing ensembles with infinite entropy are unphysical, and should be discarded.*

We now give four characterizations of the projective-measurement prediction.

Condition EQ-MEAS (Nonselective projective measurement). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x x P_x$ returns the output of a nonselective projective measurement. That is, $\Phi_X(\rho) = \sum_x P_x \rho P_x$.*

Condition EQ-LIND (Spectral equilibration). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x x P_x$ can be characterized as the equilibration under a Lindblad equation with vanishing Hamiltonian and jump operator X . That is, $\frac{d\rho}{dt} = X\rho X - \frac{1}{2}\{X^2, \rho\}$ and $\Phi_X(\rho) = \lim_{t \rightarrow \infty} \rho(t)$, where $\rho = \rho(0)$.*

Condition EQ-HS (Closest commuting ensemble). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x x P_x$ returns the ensemble commuting with X that is closest to the incoming ensemble in Hilbert–Schmidt distance. That is, $\Phi_X(\rho) = \operatorname{argmin}_{\sigma \in \mathcal{D}(\mathcal{H}), [\sigma, X]=0} \|\rho - \sigma\|_{HS}^2$.*

Condition EQ-KL (Closest commuting ensemble in information). *The spectral equilibration process Φ_X associated with an observable $X = \sum_x x P_x$ returns the ensemble commuting with X that is closest to the incoming ensemble in relative entropy. That is, $\Phi_X(\rho) = \operatorname{argmin}_{\sigma \in \mathcal{D}(\mathcal{H}), [\sigma, X]=0} D(\rho\|\sigma)$.*

3 Theorem

We now prove the equivalence.

Theorem 1. *Over $QSTO:H$, conditions [EQ-MEAS](#), [EQ-LIND](#) and [EQ-HS](#) are equivalent. Over $QSTO:H+[B]FINENT$, condition [EQ-KL](#) is also equivalent to the others.*

Proof. Let Φ_X denote the black-box spectral equilibration process associated with $X = \sum_x x P_x$, and define the pinching map $\mathcal{P}_X(\rho) = \sum_x P_x \rho P_x$. Condition [EQ-MEAS](#) states that $\Phi_X = \mathcal{P}_X$. We show directly that each of the other conditions also characterizes Φ_X as the map $\mathcal{P}_X$.

[EQ-LIND](#) $\iff$ [EQ-MEAS](#). Since the P_x are the projectors onto the eigenspaces associated with each x , we can write the density matrix as

$$\rho = \sum_{x,y} P_x \rho P_y. \quad (2)$$

This is the block-matrix decomposition of ρ with respect to the eigenspaces of X : the terms $P_x \rho P_x$ are the diagonal blocks and the terms $P_x \rho P_y$, with $x \neq y$, are the off-diagonal blocks. We now calculate the evolution of each block. Multiplying the Lindblad equation in [EQ-LIND](#) on the left by P_x and on the right by P_y gives

$$P_x \frac{d\rho}{dt} P_y = P_x X \rho X P_y - \frac{1}{2} P_x X^2 \rho P_y - \frac{1}{2} P_x \rho X^2 P_y. \quad (3)$$

Since $P_x X = x P_x$, $X P_y = y P_y$, $P_x X^2 = x^2 P_x$ and $X^2 P_y = y^2 P_y$, we have

$$\frac{d}{dt} P_x \rho(t) P_y = \left(xy - \frac{x^2}{2} - \frac{y^2}{2} \right) P_x \rho(t) P_y = -\frac{1}{2} (x - y)^2 P_x \rho(t) P_y. \quad (4)$$

Each block therefore satisfies a simple exponential-decay equation, whose solution is

$$P_x \rho(t) P_y = e^{-\frac{t}{2}(x-y)^2} P_x \rho(0) P_y. \quad (5)$$

Note that, if X is unbounded, the differential equation is initially valid only on the domain of its generator. The blockwise solution extends the resulting semigroup to every density operator. For $x = y$, the exponential is one and the diagonal block remains fixed. For $x \neq y$, the exponential tends to zero and the off-diagonal block disappears. To take the limit of all blocks at once, note that distinct blocks are orthogonal in the Hilbert–Schmidt inner product. Therefore

$$\|\rho(t) - \mathcal{P}_X(\rho(0))\|_{HS}^2 = \left\| \sum_{x \neq y} e^{-\frac{t}{2}(x-y)^2} P_x \rho(0) P_y \right\|_{HS}^2 \quad (6)$$

$$= \sum_{x \neq y} e^{-t(x-y)^2} \|P_x \rho(0) P_y\|_{HS}^2. \quad (7)$$

Moreover,

$$\sum_{x,y} \|P_x \rho(0) P_y\|_{HS}^2 = \|\rho(0)\|_{HS}^2 < \infty. \quad (8)$$

Each term in the previous sum tends to zero and is bounded by the corresponding term in this convergent sum. Dominated convergence therefore gives

$$\lim_{t \rightarrow \infty} \|\rho(t) - \mathcal{P}_X(\rho(0))\|_{HS} = 0. \quad (9)$$

Both $\rho(t)$ and $\mathcal{P}_X(\rho(0))$ are density operators. For density operators, Hilbert–Schmidt convergence to a density operator implies trace-norm convergence: this is the non-commutative Scheffé lemma. Hence

$$\lim_{t \rightarrow \infty} \|\rho(t) - \mathcal{P}_X(\rho(0))\|_1 = 0, \quad (10)$$

which means $\lim_{t \rightarrow \infty} \rho(t) = \mathcal{P}_X(\rho(0))$. Thus the Lindblad characterization returns $\mathcal{P}_X$, so [EQ-LIND](#) and [EQ-MEAS](#) characterize the same black-box process Φ_X .

[EQ-HS](#) $\iff$ [EQ-MEAS](#). An operator commutes with X if and only if it is block diagonal with respect to the eigenspaces of X . Therefore, $[\sigma, X] = 0$, σ has no off-diagonal blocks and $\mathcal{P}_X(\sigma) = \sigma$ are all equivalent statements. The difference $\rho - \mathcal{P}_X(\rho)$ contains only off-diagonal blocks, while $\mathcal{P}_X(\rho) - \sigma$ contains only diagonal blocks. They are orthogonal in the Hilbert–Schmidt inner product, and hence

$$\|\rho - \sigma\|_{HS}^2 = \|\rho - \mathcal{P}_X(\rho)\|_{HS}^2 + \|\mathcal{P}_X(\rho) - \sigma\|_{HS}^2. \quad (11)$$

The first term does not depend on σ , while the last term is non-negative and vanishes if and only if $\sigma = \mathcal{P}_X(\rho)$. Since $\mathcal{P}_X$ is positive and trace preserving, $\mathcal{P}_X(\rho)$ is itself a density operator. It is therefore the unique minimizer. Thus the Hilbert–Schmidt characterization returns $\mathcal{P}_X$, so [EQ-HS](#) and [EQ-MEAS](#) characterize the same black-box process Φ_X .

Under [\[B\]FINENT](#), [EQ-KL](#) $\iff$ [EQ-MEAS](#). Since ensembles must have finite entropy, the admissible domain of the pinching map is restricted to those density operators ρ for which both $S(\rho)$ and $S(\mathcal{P}_X(\rho))$ are finite. We show that KL minimization returns $\mathcal{P}_X$ on this same domain.

If $[\sigma, X] = 0$, then $\log \sigma$ is block diagonal. Therefore $\text{tr}(\rho \log \sigma) = \text{tr}(\mathcal{P}_X(\rho) \log \sigma)$. Moreover, $\mathcal{P}_X(\rho)$ and hence $\log \mathcal{P}_X(\rho)$ are block diagonal, so $\text{tr}(\rho \log \mathcal{P}_X(\rho)) = \text{tr}(\mathcal{P}_X(\rho) \log \mathcal{P}_X(\rho))$. We can

therefore insert and subtract this common quantity:

$$\begin{aligned}
D(\rho\|\sigma) &= \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \sigma) \\
&= \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \mathcal{P}_X(\rho)) + \text{tr}(\mathcal{P}_X(\rho) \log \mathcal{P}_X(\rho)) - \text{tr}(\mathcal{P}_X(\rho) \log \sigma) \\
&= D(\rho\|\mathcal{P}_X(\rho)) + D(\mathcal{P}_X(\rho)\|\sigma).
\end{aligned} \tag{12}$$

We also have

$$D(\rho\|\mathcal{P}_X(\rho)) = \text{tr}(\rho \log \rho) - \text{tr}(\mathcal{P}_X(\rho) \log \mathcal{P}_X(\rho)) = S(\mathcal{P}_X(\rho)) - S(\rho), \tag{13}$$

which means the KL divergence between the two ensembles is well-defined and finite if and only if $S(\mathcal{P}_X(\rho))$ is finite.

For non-faithful σ , the identity is understood on its support; candidates whose support does not contain that of ρ have infinite relative entropy. The first term does not depend on σ , while the last term is non-negative and vanishes if and only if $\sigma = \mathcal{P}_X(\rho)$. Thus the relative-entropy characterization returns $\mathcal{P}_X$, so [EQ-KL](#) and [EQ-MEAS](#) characterize the same black-box process Φ_X on ensembles of finite entropy, completing the proof. $\square$

4 Additional remarks

Equilibration as a black-box description. Condition [EQ-LIND](#) does not claim that every physical implementation of projective measurement has vanishing Hamiltonian and jump operator X . It states that the same input-output process can be described by such an equilibration. Other dynamics may have the same asymptotic map.

Relative entropy in infinite dimensions. In an infinite-dimensional Hilbert space, $D(\rho\|\mathcal{P}_X(\rho))$ can be infinite only when $S(\mathcal{P}_X(\rho)) = +\infty$. In that case, every ensemble commuting with X is infinitely distant from ρ , so the minimization in [EQ-KL](#) does not determine a unique closest ensemble. Since ensembles with infinite entropy are physically ill-defined, the real problem is the Hilbert space formalism, as it does not guarantee finite entropy for all density operators.

Observables with continuous spectra. Note that all four conditions fail when X has purely continuous spectrum, and they fail on the continuous part when both discrete and continuous spectra are present. In fact, the pinching map cannot be defined there as a sum over eigenspace projections. An operator ρ supported on a non-atomic spectral sector and commuting with X would be decomposable in the spectral representation of X and cannot be a non-zero trace-class operator. Similarly, fixed points of the Lindblad equation supported on that sector would not be density operators. In other words, there are no density-operator equilibria or equilibration processes on the continuous part of the spectrum.

5 Conclusion

Projective measurements can be characterized as black-box equilibration processes that happen at a fast scale, leaving their implementation free to match different specific experimental conditions. The repeatability of the measurement is intrinsic in requiring that all outputs are equilibria of the process. The process, then, is an equilibration towards those outcomes. As the output of the process is a classical statistical mixture of possible measurement outcomes, the second part of the

measurement (i.e. the update of the state based on the measurement outcome) is equivalent to what happens in the classical case.

More generally, the black-box idea could be extended to destructive measurements. One can imagine a black-box process in which the input is the state of the system while the output is the final state of the measurement device.